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## Unit 5 : Session 6 : SLO 2 | Symbolic Limits and Series

### I. Questions & Answers

- $\lim_{x \rightarrow \infty} 1/x = 0$  as  $x$  grows without bound,  $1/x$  shrinks to 0. In SymPy: `limit(1/x, x, oo)`  $\rightarrow 0$ .
- Sum of  $1 + 1/2 + 1/4 + \dots$  a geometric series with first term  $a = 1$  and ratio  $r = 1/2$ ;  $\text{sum} = a/(1 - r) = 1/(1 - 1/2) = 2$ .
- $\lim_{x \rightarrow 0} \sin(x)/x = 1$  the classic standard limit. In SymPy: `limit(sin(x)/x, x, 0)`  $\rightarrow 1$ .

### II. Code

- `from sympy import symbols, limit, sin, oo, summation`
- `from sympy.abc import x, n`
- `print(limit(1/x, x, oo)) # 0`
- `print(summation((1/2)**n, (n, 0, oo))) # 2`
- `print(limit(sin(x)/x, x, 0)) # 1`